Week 05 Progress Report

Details

- **Name of Student:** Tirthak Girish Likhar
- **PRN: 22070521041**
- **Name of Industry:** SS Infotech Pvt. Ltd.
- **Reporting Week: (02/01/2026 to 06/01/2026)**
- **Institute Mentor Name:** Dr. Shreyas Hole
- **Industry Mentor Name:** Mr. Viraj Patle

Overview of a Week

- During Week 5, I began moving from foundational concepts to the SIEM phase of my project. I focused on understanding how Security Information and Event Management (SIEM) tools work and why they are important in a SOC environment.

- I studied the Wazuh SIEM architecture in detail and learned about its components, such as the manager, indexer, dashboard, and endpoint agents. I also checked the system and VM requirements needed for installation and prepared the steps for setting up Wazuh in the coming week.

- Alongside this, I reviewed Linux logs again to understand how they will be forwarded and analyzed once the SIEM is fully configured. Overall, this week helped me build the necessary background and preparation for starting the actual Wazuh setup next week.

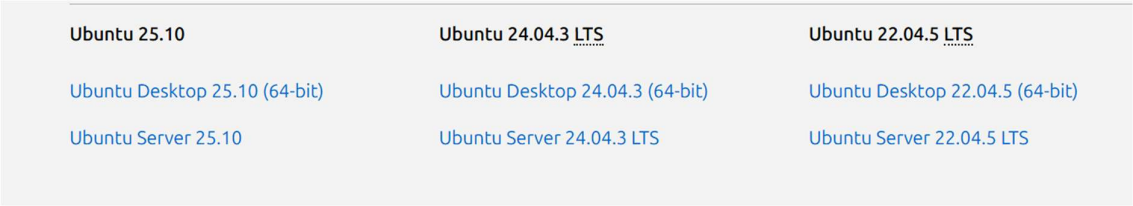
Key Accomplishments

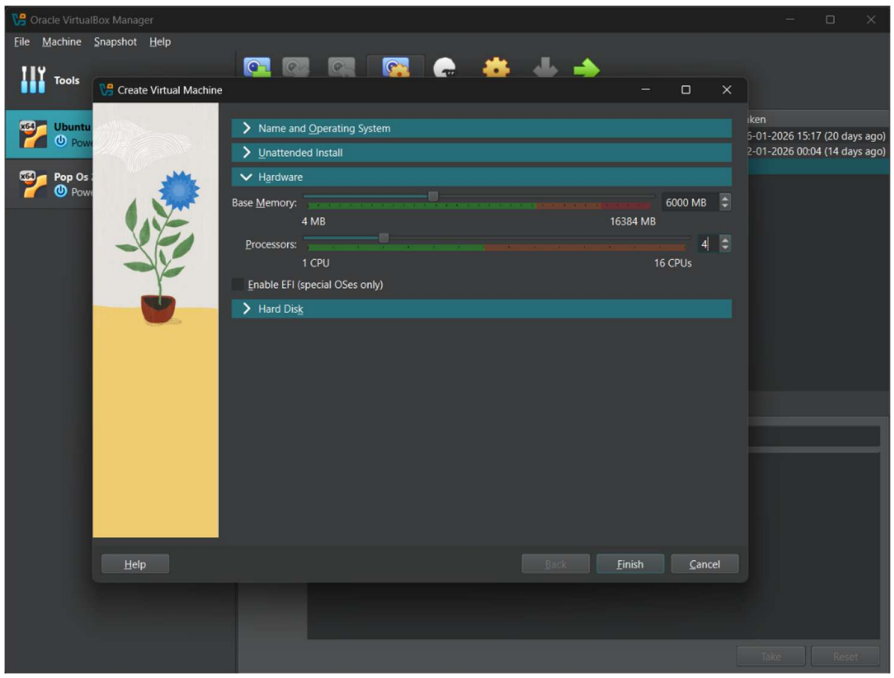
- Gained a clear understanding of SIEM concepts and how they support monitoring and incident detection in a SOC.
- Studied the complete architecture of the Wazuh SIEM, including the manager, indexer, dashboard, and agent components.
- Prepared the Ubuntu system for SIEM installation by checking resources, updating packages, and reviewing prerequisites.
- Documented the step-by-step plan for the upcoming Wazuh setup to ensure smooth installation next week.
- Reviewed important Linux logs again to understand how they will be collected and analyzed once connected to the SIEM.

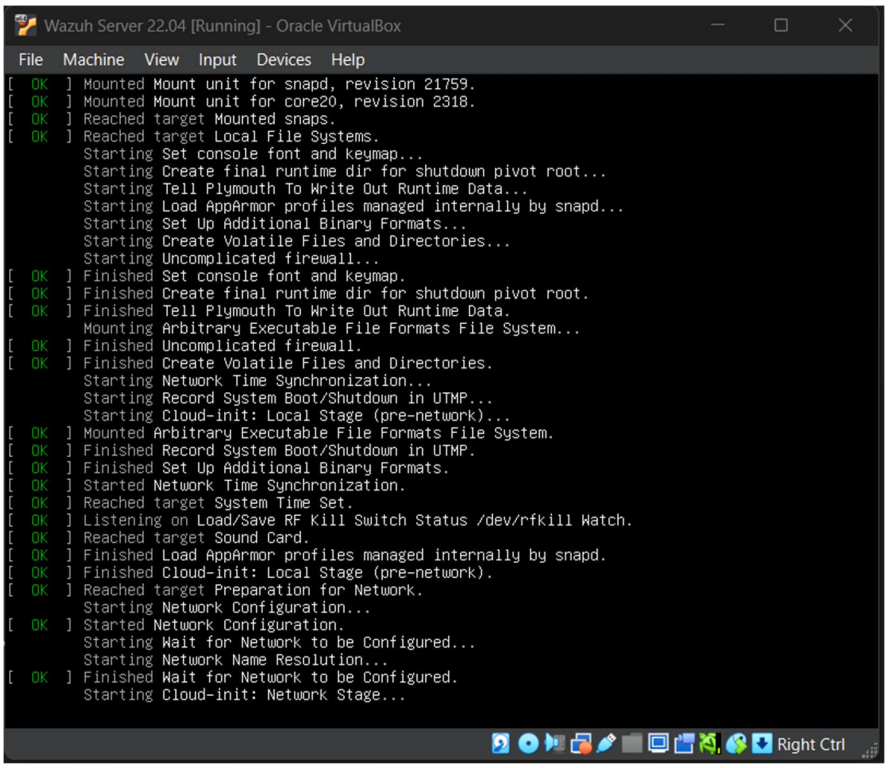
Tools Used

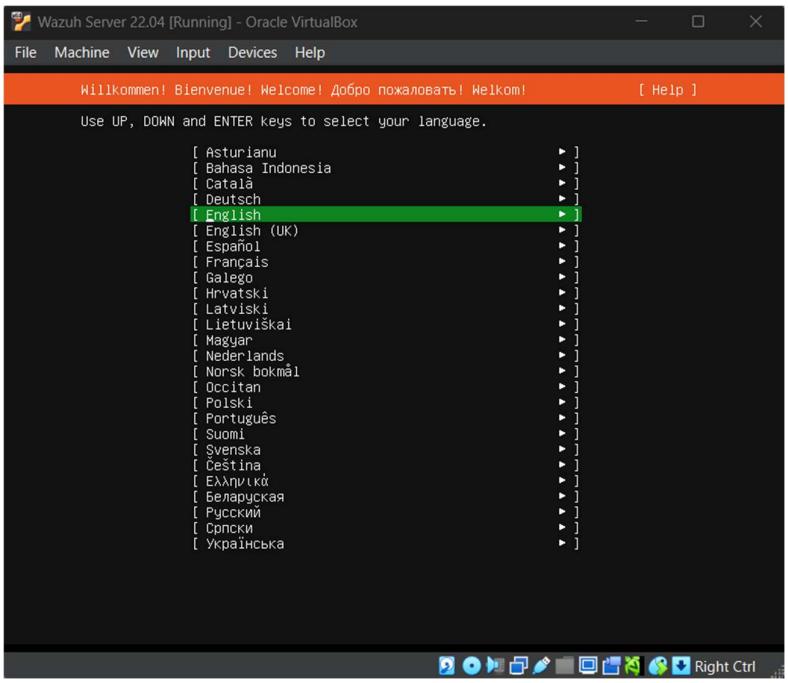
- Gained a clear understanding of SIEM concepts and how they support monitoring and incident detection in a SOC **Ubuntu 24.04 LTS (Linux VM)** – Used to prepare the system for SIEM installation.
- **VirtualBox / VMware** – For running and managing the virtual machine environment.
- **Linux Terminal & CLI Tools** – Used for system updates, package installation, and checking prerequisites.
- **Web Browser (Chrome/Firefox)** – Used for accessing Wazuh documentation and SIEM reference material.
- **Text Editor / Notes App** – For documenting installation steps and SIEM setup plan.

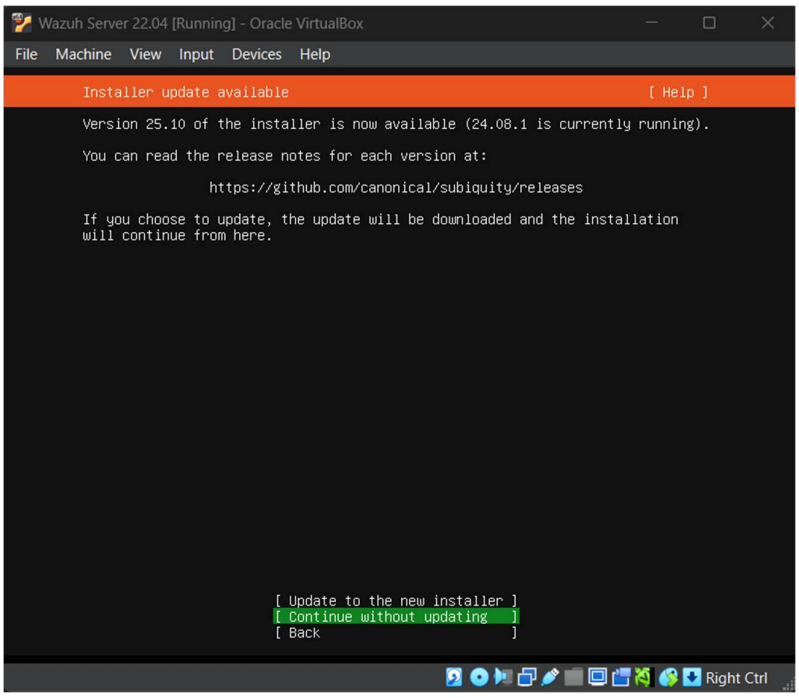
Screenshots of Activities (if Any)

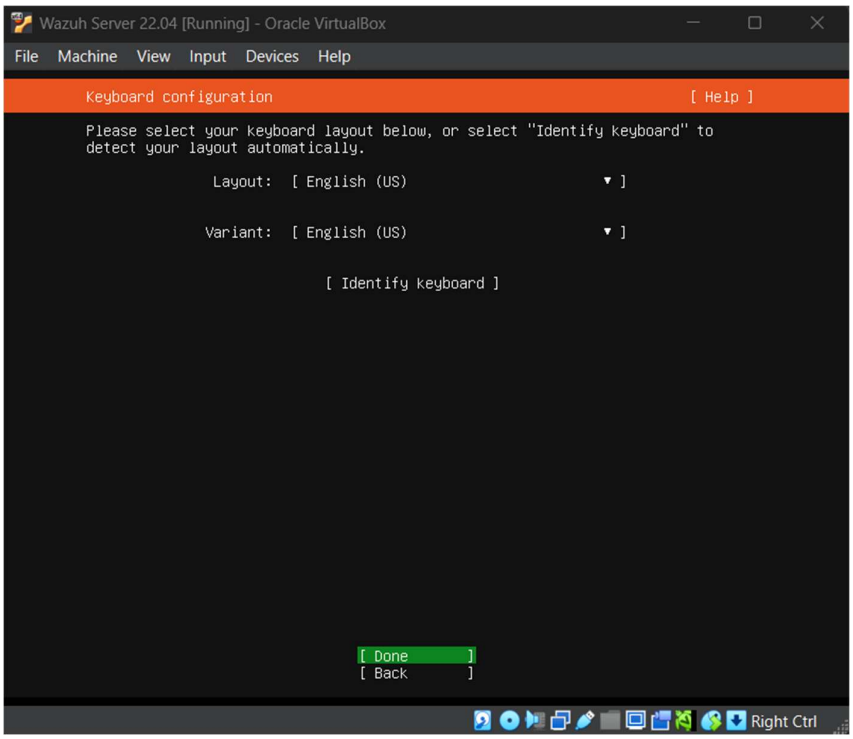


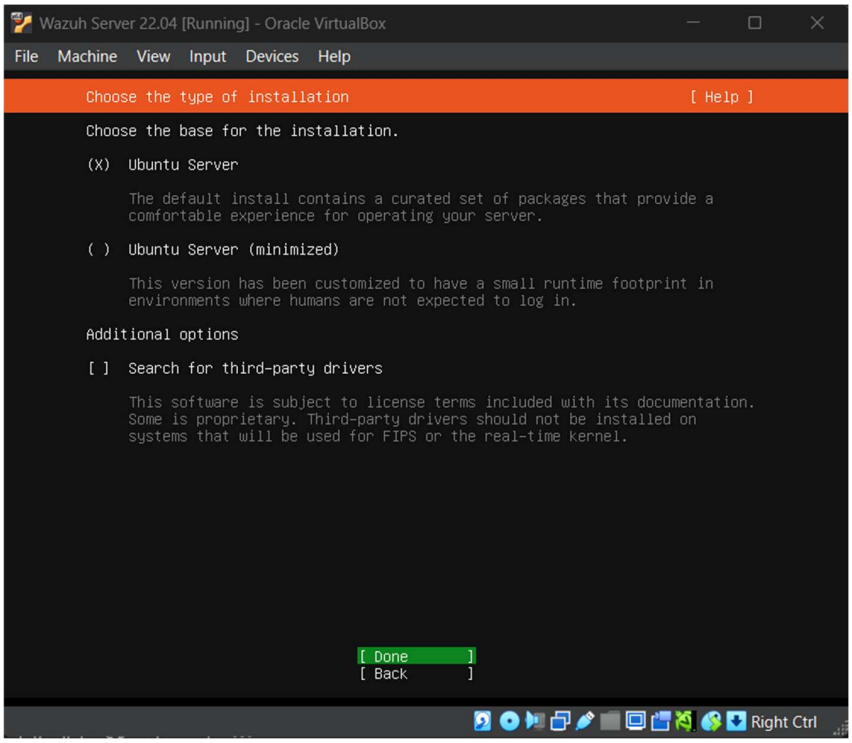

Oracle VirtualBox Manager
File Machine Snapshot Help
Tools
Ubuntu
Pop Os
Create Virtual Machine
Name and Operating System
Unattended Install
Hardware
Base Memory:
6000 MB
4 MB
16384 MB
Processors:
4
1 CPU
16 CPUs
Enable EFI (special OSes only)
Hard Disk
Help
Back
Finish
Cancel

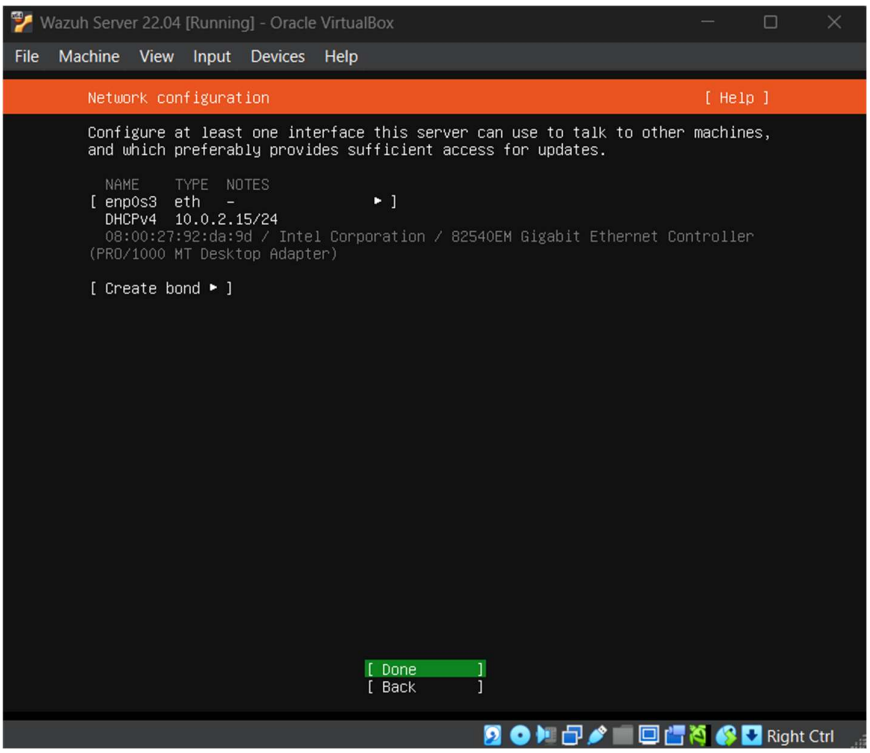

Wazuh Server 22.04 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
[OK] Mounted Mount unit for snapd, revision 21759.
[OK] Mounted Mount unit for core20, revision 2318.
[OK] Reached target Mounted snaps.
[OK] Reached target Local File Systems.
Starting Set console font and keymap...
Starting Create final runtime dir for shutdown pivot root...
Starting Tell Plymouth To Write Out Runtime Data...
Starting Load AppArmor profiles managed internally by snapd...
Starting Set Up Additional Binary Formats...
Starting Create Volatile Files and Directories...
Starting Uncomplicated firewall...
[OK] Finished Set console font and keymap.
[OK] Finished Create final runtime dir for shutdown pivot root.
[OK] Finished Tell Plymouth To Write Out Runtime Data.
Mounting Arbitrary Executable File Formats File System...
[OK] Finished Uncomplicated firewall.
[OK] Finished Create Volatile Files and Directories.
Starting Network Time Synchronization...
Starting Record System Boot/Shutdown in UTMP...
Starting Cloud-init: Local Stage (pre-network)...
[OK] Mounted Arbitrary Executable File Formats File System.
[OK] Finished Record System Boot/Shutdown in UTMP.
[OK] Finished Set Up Additional Binary Formats.
[OK] Started Network Time Synchronization.
[OK] Reached target System Time Set.
[OK] Listening on Load/Save RF Kill Switch Status /dev/rfkill Watch.
[OK] Reached target Sound Card.
[OK] Finished Load AppArmor profiles managed internally by snapd.
[OK] Finished Cloud-init: Local Stage (pre-network).
[OK] Reached target Preparation for Network.
Starting Network Configuration...
[OK] Started Network Configuration.
Starting Wait for Network to be Configured...
Starting Network Name Resolution...
[OK] Finished Wait for Network to be Configured.
Starting Cloud-init: Network Stage...
Right Ctrl

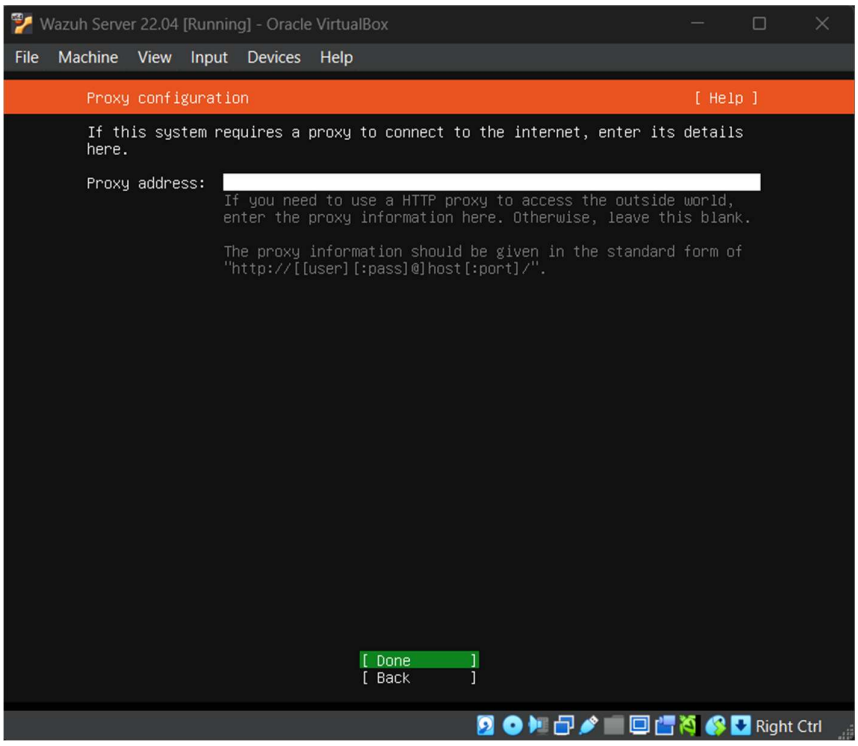

Wazuh Server 22.04 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Willkommen! Bienvenue! Welcome! Добро пожаловать! Welkom!
[Help]
Use UP, DOWN and ENTER keys to select your language.
[Asturianu]
[Bahasa Indonesia]
[Català]
[Deutsch]
[English]
[English (UK)]
[Español]
[Français]
[Galego]
[Hrvatski]
[Latviski]
[Lietuviškai]
[Magyar]
[Nederlands]
[Norsk bokmål]
[Occitan]
[Polski]
[Português]
[Suomi]
[Svenska]
[Čeština]
[Ελληνικά]
[Беларуская]
[Русский]
[Српски]
[Українська]
Right Ctrl

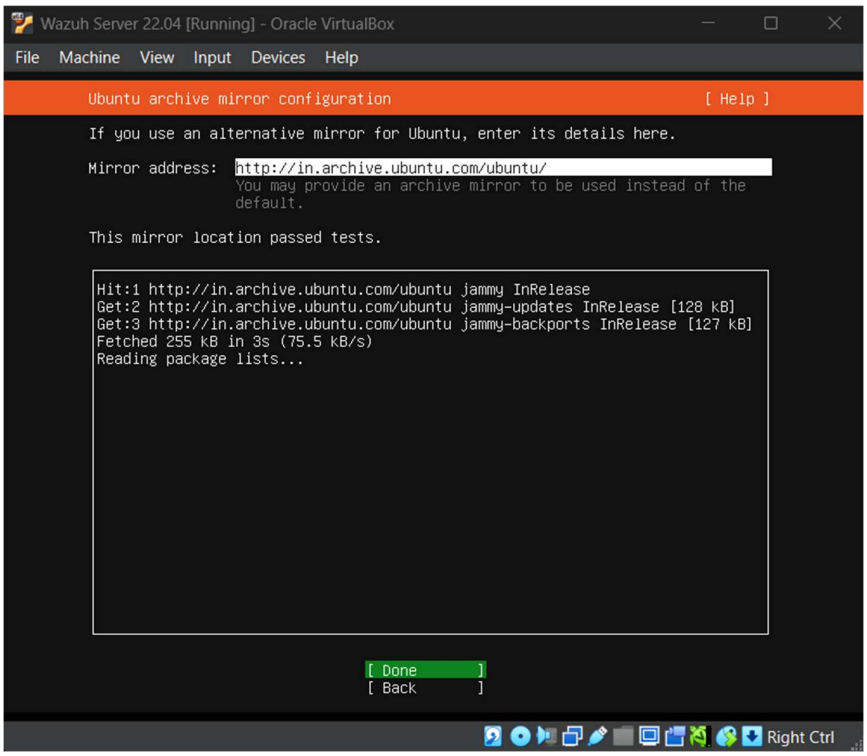

Wazuh Server 22.04 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Installer update available
[Help]
Version 25.10 of the installer is now available (24.08.1 is currently running).
You can read the release notes for each version at:
https://github.com/canonical/subiquity/releases
If you choose to update, the update will be downloaded and the installation will continue from here.
[Update to the new installer]
[Continue without updating]
[Back]
Right Ctrl

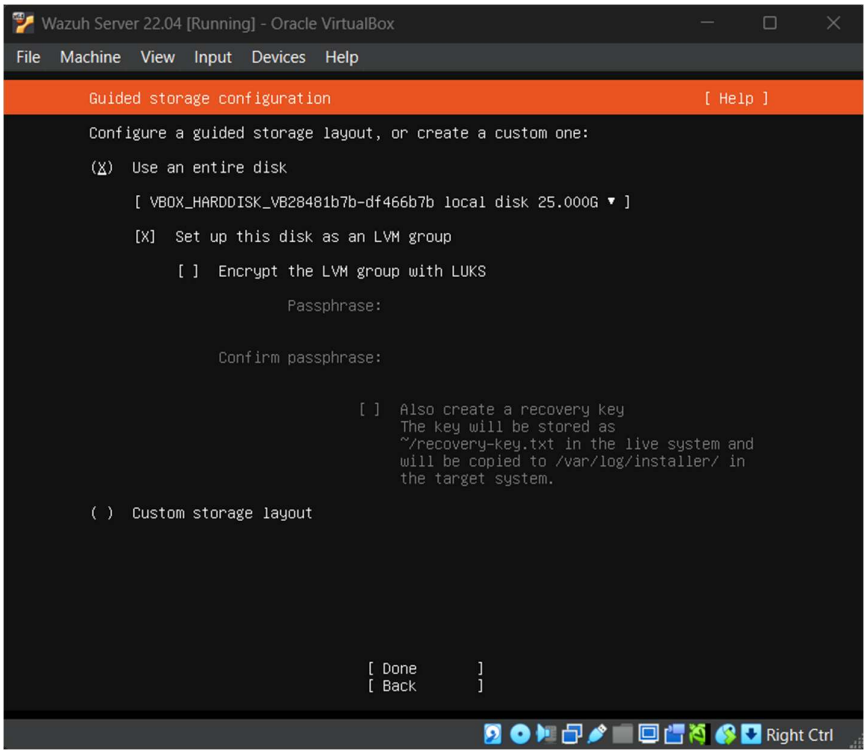

Wazuh Server 22.04 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Keyboard configuration
[Help]
Please select your keyboard layout below, or select "Identify keyboard" to detect your layout automatically.
Layout: [English (US)]
Variant: [English (US)]
[Identify keyboard]
[Done]
[Back]
Right Ctrl

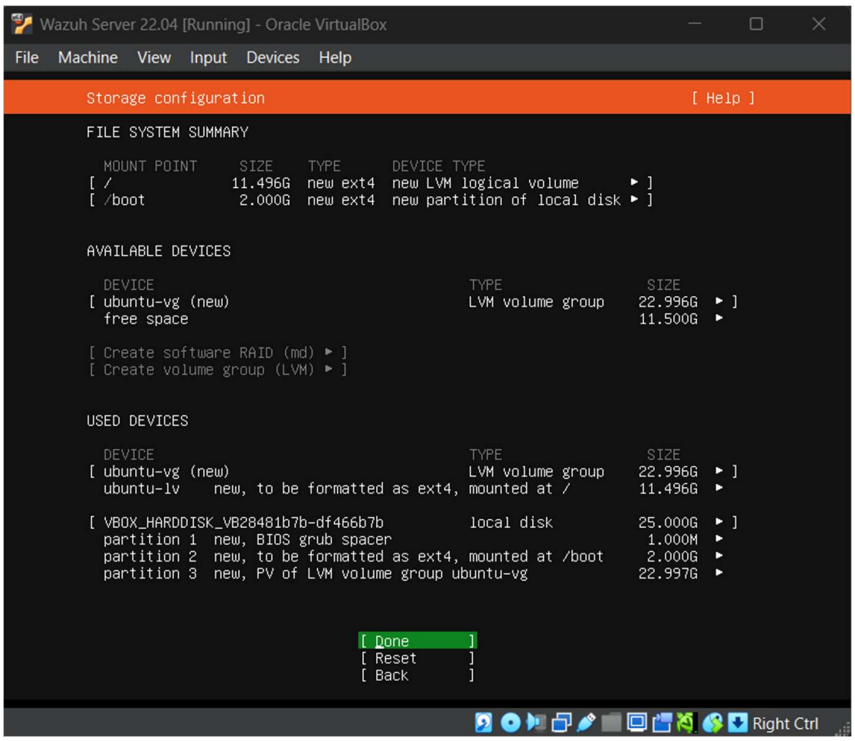

Wazuh Server 22.04 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Choose the type of installation
[Help]
Choose the base for the installation.
(X) Ubuntu Server
The default install contains a curated set of packages that provide a comfortable experience for operating your server.
( ) Ubuntu Server (minimized)
This version has been customized to have a small runtime footprint in environments where humans are not expected to log in.
Additional options
[ ] Search for third-party drivers
This software is subject to license terms included with its documentation. Some is proprietary. Third-party drivers should not be installed on systems that will be used for FIPS or the real-time kernel.
[Done]
[Back]
Right Ctrl

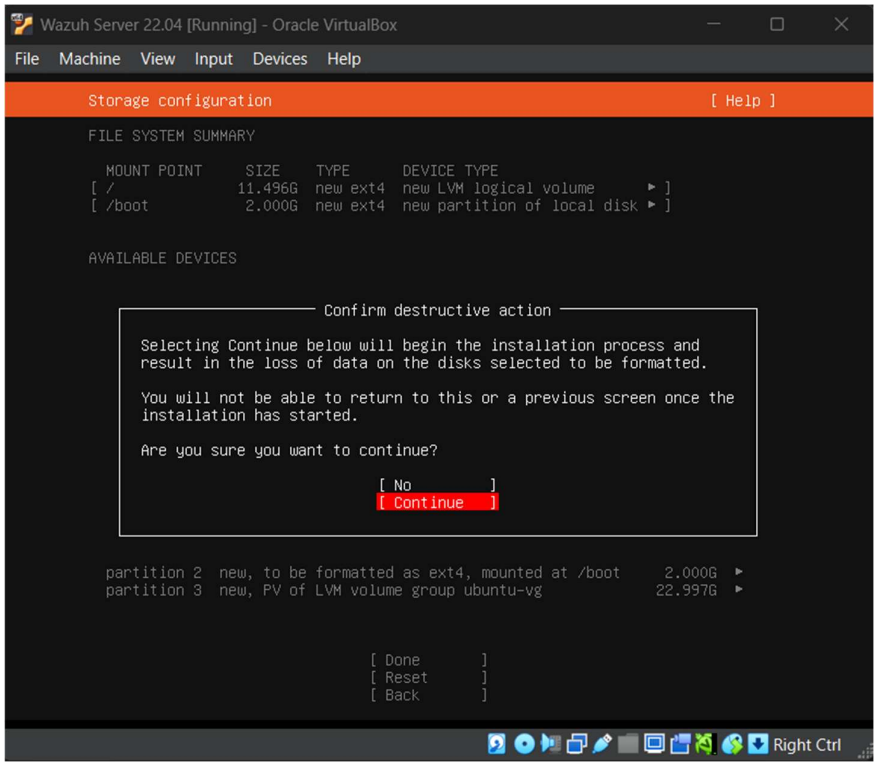

Wazuh Server 22.04 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Network configuration [Help]
Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient access for updates.
NAME TYPE NOTES
[enp0s3 eth - ▸]
DHCPv4 10.0.2.15/24
08:00:27:92:da:9d / Intel Corporation / 82540EM Gigabit Ethernet Controller (PRO/1000 MT Desktop Adapter)
[Create bond ▸]
[Done]
[Back]
Right Ctrl

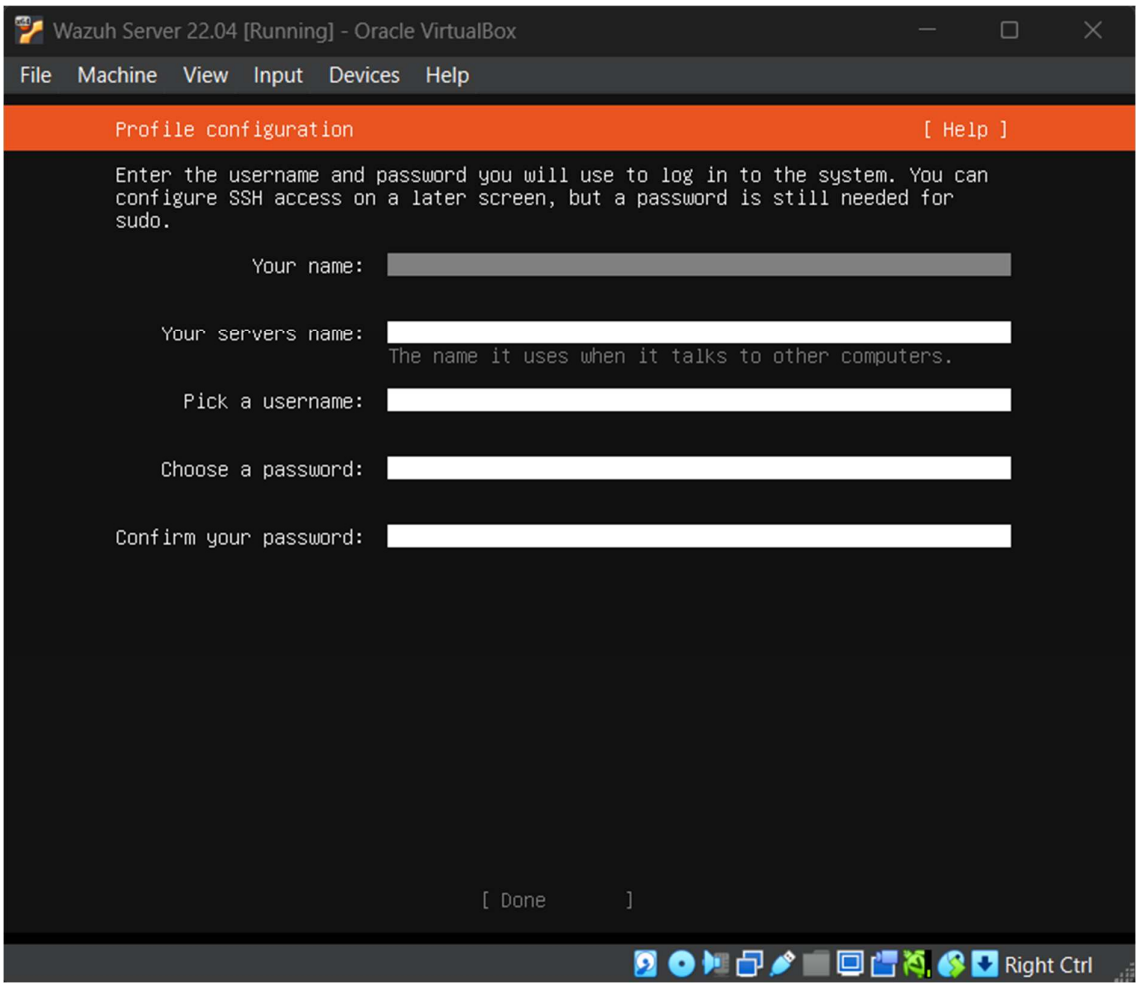

Wazuh Server 22.04 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Proxy configuration [Help]
If this system requires a proxy to connect to the internet, enter its details here.
Proxy address:
If you need to use a HTTP proxy to access the outside world, enter the proxy information here. Otherwise, leave this blank.
The proxy information should be given in the standard form of "http://[[user][:pass]@]host[:port]/".
[Done]
[Back]
Right Ctrl


Wazuh Server 22.04 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Ubuntu archive mirror configuration [Help]
If you use an alternative mirror for Ubuntu, enter its details here.
Mirror address: http://in.archive.ubuntu.com/ubuntu/
You may provide an archive mirror to be used instead of the default.
This mirror location passed tests.
Hit:1 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Fetched 255 kB in 3s (75.5 kB/s)
Reading package lists...
[Done]
[Back]
Right Ctrl


Wazuh Server 22.04 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Guided storage configuration [Help]
Configure a guided storage layout, or create a custom one:
(X) Use an entire disk
[VBOX_HARDDISK_VB28481b7b-df466b7b local disk 25.000G ▾]
[X] Set up this disk as an LVM group
[ ] Encrypt the LVM group with LUKS
Passphrase:
Confirm passphrase:
[ ] Also create a recovery key
The key will be stored as
~/recovery-key.txt in the live system and
will be copied to /var/log/installer/ in
the target system.
( ) Custom storage layout
[Done]
[Back]
Right Ctrl


Wazuh Server 22.04 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Storage configuration [Help]
FILE SYSTEM SUMMARY
MOUNT POINT SIZE TYPE DEVICE TYPE
[/ 11.496G new ext4 new LVM logical volume ▸]
[/boot 2.000G new ext4 new partition of local disk ▸]
AVAILABLE DEVICES
DEVICE TYPE SIZE
[ubuntu-vg (new) LVM volume group 22.996G ▸]
free space 11.500G ▸
[Create software RAID (md) ▸]
[Create volume group (LVM) ▸]
USED DEVICES
DEVICE TYPE SIZE
[ubuntu-vg (new) LVM volume group 22.996G ▸]
ubuntu-lv new, to be formatted as ext4, mounted at / 11.496G ▸
[VBOX_HARDDISK_VB28481b7b-df466b7b local disk 25.000G ▸]
partition 1 new, BIOS grub spacer 1.000M ▸
partition 2 new, to be formatted as ext4, mounted at /boot 2.000G ▸
partition 3 new, PV of LVM volume group ubuntu-vg 22.997G ▸
[Done]
[Reset]
[Back]
Right Ctrl


Wazuh Server 22.04 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Storage configuration [Help]
FILE SYSTEM SUMMARY
MOUNT POINT SIZE TYPE DEVICE TYPE
[/ 11.496G new ext4 new LVM logical volume ▸]
[/boot 2.000G new ext4 new partition of local disk ▸]
AVAILABLE DEVICES
Confirm destructive action
Selecting Continue below will begin the installation process and result in the loss of data on the disks selected to be formatted.
You will not be able to return to this or a previous screen once the installation has started.
Are you sure you want to continue?
[No]
[Continue]
partition 2 new, to be formatted as ext4, mounted at /boot 2.000G ▸
partition 3 new, PV of LVM volume group ubuntu-vg 22.997G ▸
[Done]
[Reset]
[Back]
Right Ctrl


Wazuh Server 22.04 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Profile configuration
[Help]
Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.
Your name:
Your servers name:
The name it uses when it talks to other computers.
Pick a username:
Choose a password:
Confirm your password:
[Done]
Right Ctrl